

Analysis of Symmetry of 2-point Correlation Functions in Finite Temperature QCD

Using Möbius Domain-Wall Fermions

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Outline

- Research Background
- Theoretical Background
- Lattice Simulation Setup
- Numerical Results
- Summary and Outlook

Research Background

- Chiral Phase Transition is a very successful way to phenomenologically describe the Hadron and Quark Gluon Plasma Structure
- The behavior of $U(1)_A$ axial symmetry restoration is important to the order of the chiral phase transition (and universality class if it is 2nd-order), so as our prediction capability with perturbative method
- The historical effective linear sigma model analysis showed the $U(1)_A$ can not be restored before the transition, if the transition is second order
- Previous Lattice simulation study, such as HotQCD, concluded that the chiral phase transition is second order and extracted the universal coefficient, while $U(1)_A$ is restored near $1.5T_c \sim 2T_c$

Motivation for this work

- However, further studies showed there's nontrivial stable IR fixed point and universality with restored $U(1)_A$ and also the possibility of weak 1st order transition.
- The previous study use Highly Improved Staggered fermion that breaks the chiral symmetry behavior in finite lattice spacing non-negligibly. So these previous conclusions of $U(1)_A$ restoration need to be careful re-examined
- The previous work in the 2-flavor case by JLQCD [Phys. Rev. D **111**, 114506 (2025)] using the Möbius Domain-Wall fermions is consistent with the scenario that both $SU(2)_L \times SU(2)_R$ chiral symmetry and $U(1)_A$ symmetry are effectively restored near $T_c \approx 165(3)$ MeV.

Main part of this work

- We extend this investigation to a more realistic $N_f = 2 + 1$ QCD case adding strange quark and improve it to finer lattice size up to $L48^3 \times T16(18)$
- We study the fixed $\beta = 4.17$ case to extract the chiral limit, and the physical point case to draw the QCD symmetry behavior in reality
- We measured spatial two point correlation function data and analysed the symmetry restoration behavior for physical point with different beta and chiral limit case by a fixed beta case with varying quark masses and lattice size
- We examine the symmetry breaking by mesons screening masses as a more phenomenologically effective observable followed the idea of work in the 2-flavor case by JLQCD [Phys. Rev. D **111**, 114506 (2025)]

Symmetry of QCD

QCD: $\mathcal{Z} = \int \mathcal{D}A \mathcal{D}\bar{q} \mathcal{D}q e^{-S}$, Gauge Boson+Fermion

$$S = \int d^4x \left[-\frac{1}{2g^2} \text{Tr} F^{\mu\nu} F_{\mu\nu} + \bar{q}_a (i \not{D} - m_a) q_a \right], \quad a \text{ in } [1, N_f]$$

Specifically, in the up-down quark mass zero limit,
Flavored chiral symmetry $SU(2)_L \times SU(2)_R$:

$$q \rightarrow \exp(i\alpha_a T^a + i\beta_a T^a \gamma_5) q, \quad \bar{q} \rightarrow \bar{q} \exp(-i\alpha_a T^a + i\beta_a T^a \gamma_5),$$

$U(1)_A$ Symmetry :

$$q \rightarrow \exp(i\epsilon \gamma_5) q, \quad \bar{q} \rightarrow \bar{q} \exp(i\epsilon \gamma_5),$$

We use the mass difference of the screening masses of different channels to estimate the degree of the symmetry breaking.

Meson Correlation Functions

We consider the following set of meson operators

$$O_{\Gamma}(x) = \bar{q}(x)\Gamma F^a q(x),$$

$$\Gamma \in \left\{ \mathbf{1}(S), \gamma_5(P), \gamma_{\mu}(V), \gamma_{\mu}\gamma_5(A), \sigma_{\mu\nu} = \frac{i}{2}[\gamma_{\mu}, \gamma_{\nu}](T), \sigma_{\mu\nu}\gamma_5(X) \right\}$$

And we measure the two-points correlation function of the meson operators of different channels. According to the symmetry properties, the asymptotic behavior of the 2-point correlation function at large distances should be approximately be $a\text{Cosh}(m_{\Gamma}(x - L/2))$

Symmetry of Correlation Functions

- Flavored chiral symmetry $SU(2)_L \times SU(2)_R$: $\delta_C O_\Gamma^a = \frac{i}{2} \beta_b \bar{q} \left(\{\Gamma, \gamma_5\} \frac{1}{2} \delta^{ab} \mathbf{1} + [\Gamma, \gamma_5] i \epsilon^{bac} \frac{\tau^c}{2} \right) q$
- $U(1)_A$ Symmetry : $\delta_A O_\Gamma^a = i \epsilon \bar{q} \{\Gamma, \gamma_5\} \frac{\tau^a}{2} q$

Γ	$[\Gamma, \gamma_5]$	$\{\Gamma, \gamma_5\}$
$\mathbf{1}(S)$		γ_5
$\gamma_5(P)$		$\mathbf{1}$
$\gamma_\mu(V)$	$\gamma_\mu \gamma_5$	
$\gamma_\mu \gamma_5(A)$	γ_μ	
$\sigma_{\mu\nu}(T)$		$\sigma_{\mu\nu} \gamma_5$
$\sigma_{\mu\nu} \gamma_5(X)$		$\sigma_{\mu\nu}$

Γ	Reference Name	Abbr.	Symmetry Correspondences
$\mathbb{I}$	Scalar	S	$\left. \begin{array}{l} U(1)_A \end{array} \right\}$
γ_5	Pseudo Scalar	PS	
γ_1, γ_2	Vector	V	$\left. \begin{array}{l} SU(2)_L \times SU(2)_R \end{array} \right\}$
$\gamma_1 \gamma_5, \gamma_2 \gamma_5$	Axial Vector	A	
$\gamma_4 \gamma_3$	Tensor	T_t	$\left. \begin{array}{l} U(1)_A \end{array} \right\}$
$\gamma_4 \gamma_3 \gamma_5$	Axial Tensor	X_t	
			$SU(2)_{CS}$

Order of Chiral Phase Transition

- Perturbative $4 - \epsilon$ expansion Renormalization Group Analysis of effective Φ^4 linear sigma model for $\langle \bar{q} P_L q \rangle$. They found for $N_f > \sqrt{3}$ flavor symmetry, there's no stable IR fixed point. [Pisarski-Wilczek (1983)]
- For $N_f \geq 3$ flavor case, the anomaly effect doesn't change the order of the transition; But for $N_f = 2$ flavor case, it becomes a quadratic term, which can smooth out the singularity and change the transition to second order.
- If the anomaly effect still exists, the transition can be $O(4)$ class.
- But recently higher-loop computations and FRG computations suggested different possibilities [E.Vicari ,F.Basile, A.Pelissetto 2005 , G. Fejos 2022], and Conformal Bootstrap results indicate some interesting points too[Nakayama Ohtsuki 2016]

Possible Emergent Larger Symmetry

- Perturbative analysis of screening masses from NRQCD₃ [Laine & Vepsäläinen (2004)], solving the equation for the $N_f = 2$ case yields:

$$m_{\text{scr}} = 2\pi T + 0.1872g^2T$$

- It predicts the flavor non-singlet channel will tend to behave same screening mass. Thus, a larger symmetry, Chiral-Spin Symmetry $SU(2)_{CS}$, may emerge

$$q \rightarrow \exp(i\Sigma^i \epsilon_i)q,$$

$$\bar{q} \rightarrow \bar{q}\gamma_0 \exp(-i\Sigma^i \epsilon_i)\gamma_0,$$

where Σ^i has three components,

$$\Sigma^i = (\gamma_k, -i\gamma_5\gamma_k, \gamma_5), \quad k = 1, 2.$$

Domain-Wall Fermions (DWF)

- Overlap Dirac Operator:

The massless operator $D_{\text{ov}}(0) = \frac{1}{2}(1 + \gamma_5 \text{sgn}(H_W))$ with Hermitian H_W , satisfies the Ginsparg-Wilson relation:

$$\gamma_5 D_{\text{ov}} + D_{\text{ov}} \gamma_5 = 2 D_{\text{ov}} \gamma_5 D_{\text{ov}}$$

The massive overlap operator is defined as:

$$D_{\text{ov}}(m) = D_{\text{ov}}(0) + m(1 - D_{\text{ov}}(0)) = \frac{1+m}{2} + \frac{1-m}{2} \gamma_5 \text{sgn}(H_W)$$

- 5D Domain-Wall Realization:

DWF approximates the sign function $\text{sgn}(H_W)$ by introducing a 5th dimension of size L_s .

The effective 4D operator is:

$$D_{\text{DW}}^{4\text{D}}(m) = \frac{1+m}{2} + \frac{1-m}{2} \gamma_5 \tanh(L_s \tanh^{-1}(H_M))$$

It converges exponentially to the exact overlap operator $D_{\text{ov}}(m)$.

Lattice QCD Numerical Setup

5D Möbius Domain Wall Dirac Operator ($L_s = 12$):

$$D_{s,s'}^{5D}(m) = D_+^{(s)}\delta_{s,s'} + D_-^{(s)}(P_+\delta_{s-1,s'} + P_-\delta_{s+1,s'}) - m(D_-^{(1)}P_+\delta_{s,1}\delta_{s',L_s} + D_-^{(L_s)}P_-\delta_{s,L_s}\delta_{s',1})$$

where $D_+^{(s)} = b_5 D_W + 1$, $D_-^{(s)} = c_5 D_W - 1$, and $P_\pm = \frac{1 \pm \gamma_5}{2}$.

It corresponds to $H_{\text{MDWF}} = \frac{\gamma_5(b+c)D_W}{2+(b-c)D_W}$, and we use $b = 1.5$, $c = 0.5$.

Residual Mass m_{res} and Chiral Breaking Δ_L :

$$\bar{\psi}\Delta_L\psi \simeq m_{\text{res}}\bar{\psi}\psi + O(\psi, \bar{\psi}, A) + \mathcal{O}(a^2), \quad am_{\text{res}} = \frac{\sum_{\vec{x}} \langle J_{5q}(\vec{x}, t) P(\vec{0}, 0) \rangle}{\sum_{\vec{x}} \langle P(\vec{x}, t) P(\vec{0}, 0) \rangle}$$

It works very well at high beta case:

T (MeV)	138	145	153	157	164.5	176	202	241.6
m_{res}/m_l	0.9087	0.5360	0.3394	0.2744	0.1836	0.1041	0.0299	0.0049
m_{res}	7.31e-4	4.98e-4	3.40e-4	2.80e-4	1.91e-4	1.08e-4	2.81e-5	3.75e-6

Lattice Setup & Configurations at Physical Point

- The simulation is run with lattice size $48^3 \times 16$ near the physical point.
- To improve data quality, we performed 16-source measurements for all channels and directions.
- The simulation parameters (left) and total configuration count N_{conf} (right) are summarized below:

β	T (MeV)	am_s	am
4.13	138	0.043547	0.000805
4.15	145	0.040843	0.000930
4.17	153	0.038400	0.001001
4.18	157	0.037265	0.001022
4.20	164.5	0.035150	0.001041
4.23	176	0.032315	0.001033
4.30	202	0.026930	0.000939

β	4.13	4.15	4.17	4.18	4.20	4.23	4.30
T (MeV)	138	145	153	157	164.5	176	202
N_{conf}	395	398	396	402	407	395	404
Total Meas.	6320	6368	6336	6432	6512	6320	6464

Lattice Setup at Fixed $\beta = 4.17$

To study the symmetry restoration mechanism in the chiral limit, we perform simulations at a fixed coupling $\beta = 4.17$, varying the bare light quark mass am_l and the temporal lattice size N_t :

Lattice Size	am_s	am_l	T (MeV)
$32^3 \times 12$	0.040	0.0020, 0.0035, 0.0070, 0.0120	220.00
$32^3 \times 14$	0.040	0.0020, 0.0035, 0.0070, 0.0120	188.57
$32^3 \times 16$	0.040	0.0020, 0.0035, 0.0070, 0.0120	165.00
$36^3 \times 18$	0.040	0.0020, 0.0035, 0.0070, 0.0120	146.67
$40^3 \times 16$	0.040	0.0020, 0.0035, 0.0070, 0.0120	165.00
$48^3 \times 18$	0.040	0.0020, 0.0035, 0.0070, 0.0120	146.67

Meson Channels & Directional Averaging

- To improve the data quality, the correlation functions of each meson channel are averaged over all equivalent spatial directions and components.
- The directional averaging components is summarized below:

Channel	Components	Averaged Directions (configured in <code>config.py</code>)
AV (Axial Vector)	6	X (AVector2, AVector3), Y (AVector1, AVector3), Z (AVector1, AVector2)
Vec (Vector)	6	X (Vector2, Vector3), Y (Vector1, Vector3), Z (Vector1, Vector2)
S (Scalar)	3	X, Y, Z (TVector4)
PS (Pseudo-Scalar)	3	X, Y, Z (TAVector4)
Tt (Tensor)	3	X (TVector1), Y (TVector2), Z (TVector3)
Xt (Axial Tensor)	3	X (TAVector1), Y (TAVector2), Z (TAVector3)

Extracting Screening Masses: Methodology

- Local Effective Mass: Numerically solve the ratio at each spatial coordinate x to obtain the local effective mass $m_{\text{eff}}(x)$:

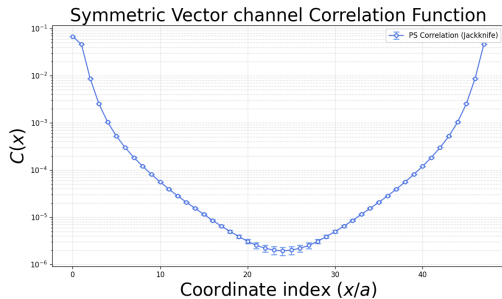
$$\frac{C(x)}{C(x+1)} = \frac{\cosh(m_{\text{eff}}(x)(x - L/2))}{\cosh(m_{\text{eff}}(x)(x+1 - L/2))}$$

- Plateau and Range: The plateau of $m_{\text{eff}}(x)$ at large separations is used to determine a suitable fitting range $[x_{\min}, x_{\max}]$.
- Screening Mass Fit: The physical screening mass m_{scr} is extracted by performing a correlated χ^2 fit of $C(x)$ to the asymptotic form:

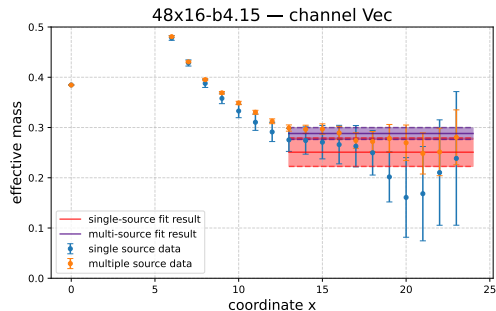
$$C(x) = A \cosh \left(m_{\text{scr}} \left(x - \frac{L}{2} \right) \right)$$

- Statistical Error: Jackknife method and Binning analysis are used to handle statistical uncertainties and eliminate HMC autocorrelation bias.

Extracting Screening Masses: Fit Examples

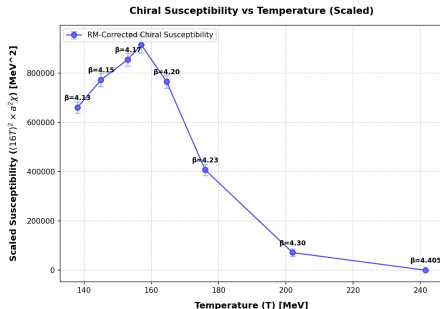


correlation data of Vec channel for
 $\beta = 4.15 (T=153\text{MeV})$



effective mass of Vec channel for
 $\beta = 4.15 (T=153\text{MeV})$

Disconnected Chiral Susceptibility



Disconnected susceptibility χ_{dis} vs Temperature

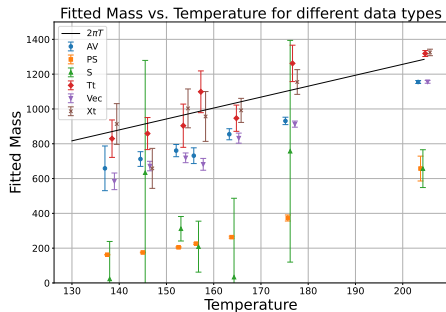
- Definition:

$$a^2 \chi_{\text{dis}} = \frac{1}{N_s^3 N_\tau} \left(\frac{N_f}{4} \right)^2 \cdot \left(\langle (\text{Tr} \not{D}^{-1})^2 \rangle - \langle \text{Tr} \not{D}^{-1} \rangle^2 \right)$$

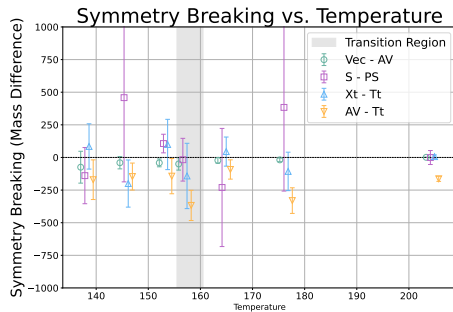
- Fluctuations of order parameter, related to topological effects.
- Clear peak determines pseudo-critical temperature: $T_c \approx 158$ MeV.

Numerical Results

Multi-source analysis results (grey band shows 2-flavor $T_c = 158(3)$ MeV range, while for chiral limit I draw $T_c \approx 165(3)$ MeV [PRD 111, 114506 (2025)]):



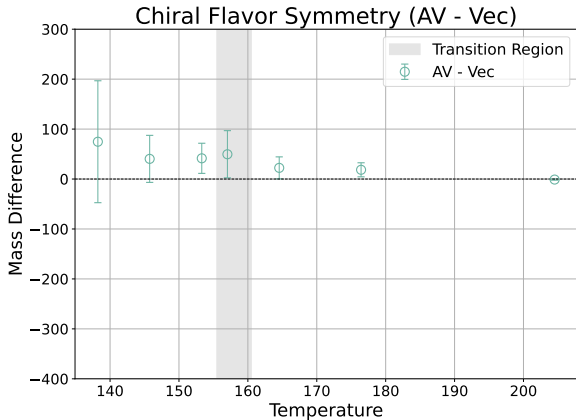
simulated effective masses of different channels



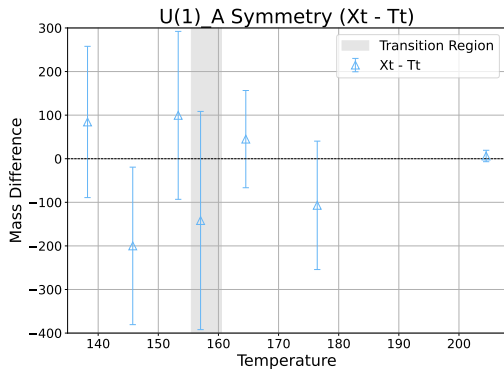
Symmetry breaking of different channels

$SU(2)_L \times SU(2)_R$ Chiral Symmetry

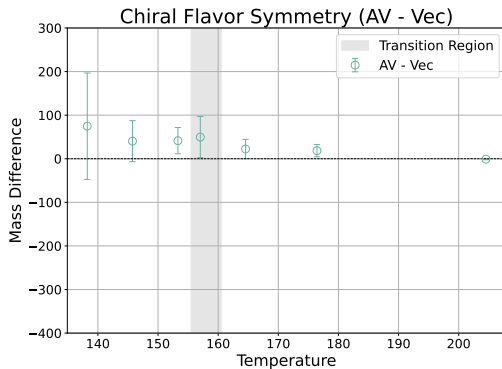
- This indicates the restoration of the $SU(2)_L \times SU(2)_R$ chiral symmetry (e.g., AV vs Vec channels).
- At $T = 0$, the difference between a_1 (1230 MeV) and ρ meson (775 MeV) is around 455 MeV.
- But the pseudo critical temperature is not very clear (due to finite quark mass)



$U(1)_A$ Symmetry ($X_t - T_t$)



Axial Tensor vs Tensor ($U(1)_A$)

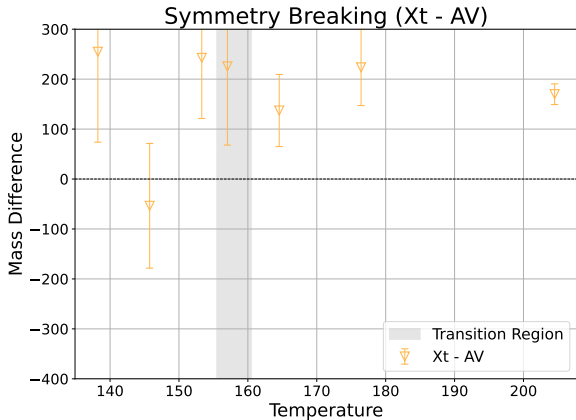


Vector vs Axial-Vector (Flavor Symmetry)

$SU(2)_{CS}$ Symmetry

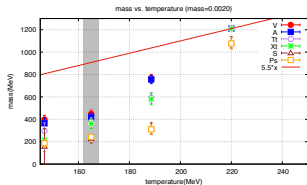
Observation: Chiral-Spin (CS) symmetry is not yet completely visible at the temperatures analyzed near the phase transition.

Requires exploration of higher temperature domains.

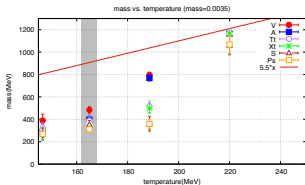


Screening Masses at Fixed $\beta = 4.17$

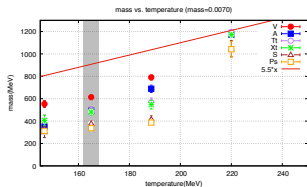
Temperature dependence of screening masses for different light quark masses:



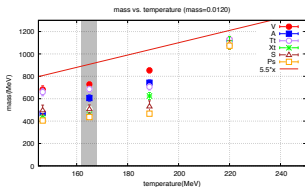
$$am_l = 0.0020$$



$$am_l = 0.0035$$



$$am_l = 0.0070$$

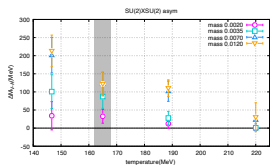


$$am_l = 0.0120$$

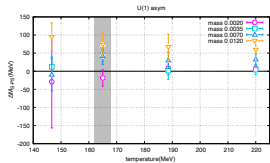
- Screening masses plotted as a function of temperature.
- As temperature increases, partner channels in PS-S and V-AV sectors become degenerate.
- This degeneration is observed for all simulated light quark masses.

Symmetry Restoration at Fixed $\beta = 4.17$

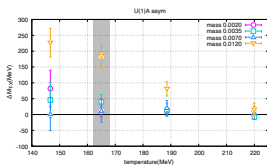
Asymmetry observables as a function of temperature for various light quark masses:



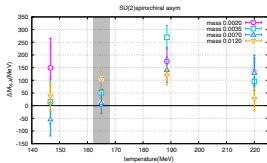
$SU(2)_L \times SU(2)_R$ (V-AV)



$U(1)$ (S-PS)



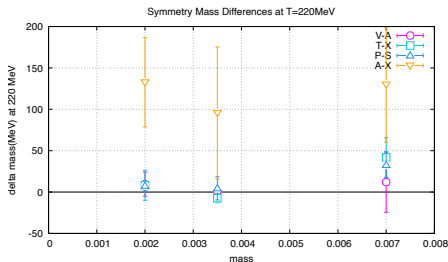
$U(1)_A$ (T-X)



$SU(2)_{CS}$ (X-AV)

- All symmetries show a clear trend towards restoration above T_c .
- $U(1)_A$ and chiral $SU(2)_L \times SU(2)_R$ symmetries show an almost simultaneous restoration as we approach the chiral limit.
- Chiral-spin symmetry $SU(2)_{CS}$ remains broken in the analyzed temperature range.

Chiral Limit Extrapolation ($T \approx 220$ MeV)



Mass differences between mesonic partners as a function of am_l at $T \approx 220$ MeV

- Analyzed mass differences at a high-temperature snapshot ($T \approx 220$ MeV) as a function of the bare light quark mass am_l .
- As $am_l \rightarrow 0$, the mass differences between chiral partners ($m_{\text{Vec}} - m_{\text{AV}}$ and $m_{\text{S}} - m_{\text{PS}}$) extrapolate toward zero.
- This supports the scenario of a **simultaneous restoration** of chiral flavor and axial $U(1)_A$ symmetries in the chiral limit.

Summary and Outlook

- Physical Point: Crossover transition with $T_c \approx 158$ MeV. Above T_c , chiral and $U(1)_A$ symmetry breakings are consistent with zero within errors.
- Chiral Limit Study: Extrapolations to $am_l \rightarrow 0$ support simultaneous restoration of chiral flavor and $U(1)_A$ symmetries (with $T_c \approx 165$ MeV).
- Chiral-Spin Symmetry: Remains broken up to 220 MeV, indicating active non-perturbative chromomagnetic interactions in QGP.
- Transition Scenario:
 - Previous: 1st-order if $U(1)_A$ restored; 2nd-order $O(4)$ if $U(1)_A$ remains broken.
 - New: $O(4)$ if $U(1)_A$ remains; weak 1st-order, or 2nd-order ($O(4) \times Z_2$, $O(4) \times Z_4$, $O(4) \times O(2)$) if $U(1)_A$ restored.

The symmetry restoration behavior of $U(1)_A$ and flavor $SU(2)_L \times SU(2)_R$ will determine the final universality class